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ELECTRONIC PACKAGE AND MANUFACTURING METHOD THEREOF

Abstract

An electronic package is provided and includes an electronic structure and a plurality of conductive pillars embedded in a cladding layer, a circuit structure formed on the cladding layer, and a reinforcing member bonded to a side surface of the cladding layer, where a plurality of electronic elements are disposed on and electrically connected to the circuit structure, such that the electronic structure electrically bridges any two of the electronic elements via the circuit structure, so as to enhance the structural strength of the electronic package and avoid warpage by means of the design of the reinforcing member.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation of U.S. patent application Ser. No. 17/950,914 filed Sep. 22, 2022, which claims benefit of priority to Taiwanese Patent Application No. 111127968 filed Jul. 26, 2022, the entire contents of which are incorporated herein by reference.

BACKGROUND
1. Technical Field
[0002] The present disclosure relates to a semiconductor device, and more particularly, to an electronic package with a bridge die and a manufacturing method thereof.
2. Description of Related Art
[0003] With the vigorous development of the electronic industry, electronic products are also gradually moving towards the trend of multi-function and high performance. Meanwhile, the technologies currently applied in the field of chip packaging include, for example, Chip Scale Package (CSP), Direct Chip Attached (DCA) or Multi-Chip Module (MCM) and other flip-chip packaging modules, etc.
[0004] FIG. **1** is a schematic cross-sectional view of a conventional semiconductor package **1**. The semiconductor package **1** includes: a cladding layer **15**, a bridge chip **10** (which has a plurality of electrode pads **100**) and a plurality of conductive pillars **13** embedded in the cladding layer **15**, a first routing structure **11** disposed on an upper side **15***a* of the cladding layer **15** and electrically connected to the bridge chip **10** and the plurality of conductive pillars **13**, a plurality of electronic elements **16** disposed on the first routing structure **11**, a package layer **18** for covering the plurality of electronic elements **16**, a second routing structure **12** disposed on a lower side **15***b* of the cladding layer **15** and electrically connected to the plurality of conductive pillars **13**, and a plurality of conductive elements **17** disposed on the second routing structure **12** and electrically connected to the second routing structure **12**.
[0005] The first routing structure **11** includes a plurality of insulating layers **110**, a plurality of routing layers **111** disposed on the plurality of insulating layers **110**, and a plurality of conductive blind vias **112** electrically connected to the routing layers **111**, so that the plurality of conductive blind vias **112** are electrically connected to the plurality of conductive pillars **13** and the plurality of routing layers **111**, and the outermost routing layer **111** has a plurality of electrical contact pads **113** of micro pad (μ-pad) specification.
[0006] The electronic element **16** is a functional chip, which has a plurality of electrode pads **160**

bonded with conductive bumps **161** such as micro bump (μ-bump) specifications, so that the electronic element **16** is soldered on the electrical contact pads **113** with the conductive bumps **161** by a flip-chip method, and then the conductive bumps **161** are covered with an underfill **162**.

[0007] However, in the conventional semiconductor package **1**, as the functional requirements of the product increase greatly, the functional requirements of the electronic element **16** also increase accordingly, so the circuit layout area of the first routing structure **11** and the second routing structure **12** increases accordingly. In this case, the width of the overall layout increases, resulting in extremely weak structural strength, which is easily affected by high temperature and causes warpage, thereby causing the routing layers **111** or the circuits of the second routing structure **12** to be broken.

[0008] Therefore, how to overcome the above-mentioned drawbacks of the prior art has become an urgent issue to be solved at present.

SUMMARY

[0009] In view of the various deficiencies of the prior art, the present disclosure provides an electronic package, comprising: a cladding layer having a first surface and a second surface opposing the first surface; an electronic structure embedded in the cladding layer; a plurality of conductive pillars embedded in the cladding layer; at least one reinforcing member bonded to the cladding layer; a circuit structure formed on the first surface of the cladding layer and electrically connected to the plurality of conductive pillars and the electronic structure, wherein the circuit structure is free from being electrically connected to the reinforcing member; and a plurality of electronic elements disposed on and electrically connected to the circuit structure, wherein the electronic structure acts as a bridge die to electrically bridge at least two of the plurality of electronic elements via the circuit structure.

[0010] The present disclosure also provides a method of manufacturing an electronic package, comprising: forming a plurality of conductive pillars on a carrier and disposing at least one reinforcing member and an electronic structure on the carrier; forming a cladding layer on the carrier, wherein the cladding layer covers the plurality of conductive pillars, the reinforcing member and the electronic structure; forming a circuit structure on the cladding layer, wherein the circuit structure is electrically connected to the plurality of conductive pillars and the electronic structure and is free from being electrically connected to the reinforcing member; disposing a plurality of electronic elements on the circuit structure, wherein the plurality of electronic elements are electrically connected to the circuit structure, such that the electronic structure acts as a bridge die to electrically bridge at least two of the plurality of electronic elements via the circuit structure; and removing the carrier.

[0011] In the aforementioned electronic package and the manufacturing method thereof, the reinforcing member is a metal frame or a frame made of non-metallic material.

[0012] In the aforementioned electronic package and the manufacturing method thereof, the reinforcing member has a width of at least 35 micrometers.

[0013] In the aforementioned electronic package and the manufacturing method thereof, the reinforcing member has a width greater than or equal to a width of each of the conductive pillars.

[0014] In the aforementioned electronic package and the manufacturing method thereof, the reinforcing member is annular.

[0015] In the aforementioned electronic package and the manufacturing method thereof, the reinforcing member surrounds the electronic structure.

[0016] In the aforementioned electronic package and the manufacturing method thereof, the reinforcing member has at least one guide hole.

[0017] In the aforementioned electronic package and the manufacturing method thereof, the electronic structure is defined with a horizontal center line to divide the cladding layer into a first area and a second area opposing the first area, such that a number of the conductive pillars in the first area is greater than a number of the conductive pillars in the second area, and a width of the

reinforcing member corresponding to the second area is greater than a width of the reinforcing member corresponding to the first area.
[0018] In the aforementioned electronic package and the manufacturing method thereof, the present disclosure further comprises forming a plurality of conductive elements on the cladding layer, wherein the plurality of conductive elements are electrically connected to the plurality of conductive pillars and are free from being electrically connected to the reinforcing member.
[0019] As can be seen from the above, the electronic package of the present disclosure and the manufacturing method thereof are designed to enhance the structural strength of the electronic package by means of the design of the reinforcing member. Therefore, compared with the prior art, the present disclosure can effectively avoid the warping problem of the electronic package.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0020] FIG. **1** is a schematic cross-sectional view of a conventional semiconductor package.
[0021] FIG. **2**A, FIG. **2**B-**1**, FIG. **2**C-**1**, FIG. **2**D, FIG. **2**E, FIG. **2**F, FIG. **2**G and FIG. **2**H-**1** are schematic cross-sectional views illustrating a manufacturing method of an electronic package according to the present disclosure.
[0022] FIG. **2**B-**2** is a schematic partial top view of FIG. **2**B-**1**.
[0023] FIG. **2**C-**2** is a schematic partial top view of FIG. **2**C-**1**.
[0024] FIG. **2**H-**2** is a schematic cross-sectional view of another aspect of FIG. **2**H-**1**.
[0025] FIG. **3** is a schematic cross-sectional view of the subsequent process of FIG. **2**H-**1**.
[0026] FIG. **4**A is a schematic cross-sectional view of another embodiment of the process of FIG. **2**B-**1**.
[0027] FIG. **4**B is a schematic partial perspective view of FIG. **4**A.
[0028] FIG. **4**C is a schematic plan view of another aspect of FIG. **4**B.
[0029] FIG. **4**D is a schematic perspective view of another aspect of FIG. **4**B.
DETAILED DESCRIPTIONS
[0030] The following describes the implementation of the present disclosure with examples. Those skilled in the art can easily understand other advantages and effects of the present disclosure from the contents disclosed in this specification.
[0031] It should be understood that, the structures, ratios, sizes, and the like in the accompanying figures are used for illustrative purposes to facilitate the perusal and in comprehension of the contents disclosed in the present specification by one skilled in the art, rather than to limit the conditions for practicing the present disclosure. Any modification of the structures, alteration of the ratio relationships, or adjustment of the sizes without affecting the possible effects and achievable proposes should still be deemed as falling within the scope defined by the technical contents disclosed in the present specification. Meanwhile, terms such as “upper,” “first,” “second,” “one,” “a,” “an” and the like used herein are merely used for clear explanation rather than limiting the practicable scope of the present disclosure, and thus, alterations or adjustments of the relative relationships thereof without essentially altering the technical contents should still be considered in the practicable scope of the present disclosure.
[0032] FIG. **2**A, FIG. **2**B-**1**, FIG. **2**C-**1**, FIG. **2**D, FIG. **2**E, FIG. **2**F, FIG. **2**G and FIG. **2**H-**1** are schematic cross-sectional views illustrating a manufacturing method of an electronic package **2** according to the present disclosure.
[0033] As shown in FIG. **2**A, an electronic structure **2**_a_ and a carrier **9** having an insulating layer **24** disposed thereon are provided, and a plurality of conductive pillars **23** are formed on the carrier **9**.
[0034] In an embodiment, the carrier **9** is, for example, a plate of semiconductor material (such as

silicon or glass), on which a release layer **90** and a metal layer **91** such as titanium/copper are sequentially formed by, for example, coating, so that the insulating layer **24** is formed on the metal layer **91**. For example, the material for forming the insulating layer **24** is a dielectric material such as polybenzoxazole (PBO), polyimide (PI), prepreg (PP), or other dielectric materials.
[0035] Furthermore, the material for forming the plurality of conductive pillars **23** is
[0036] a metal material such as copper or a solder material, and the plurality of conductive pillars **23** extend through the insulating layer **24** to contact the metal layer **91**. For example, by exposure and development, a plurality of openings exposing the metal layer **91** are formed on the insulating layer **24**, so that the plurality of conductive pillars **23** are formed by electroplating the metal layer **91** from the openings.
[0037] The electronic structure **2***a* includes an electronic body **21**, a circuit portion **22**, a plurality of first conductors **21***a* formed on the electronic body **21**, and a plurality of second conductors **22***a* formed on the circuit portion **22** and electrically connected to the circuit portion **22**. Next, a first protective layer **21***b* is formed on the electronic body **21** so that the first protective layer **21***b* covers the plurality of first conductors **21***a*, and a second protective layer **22***b* is formed on the circuit portion **22** so that the second protective layer **22***b* covers the plurality of second conductors **22***a*.
[0038] In an embodiment, the electronic body **21** is a silicon substrate, such as a semiconductor chip, which has a plurality of conductive vias **210** such as conductive through-silicon vias (TSVs) penetrating through the electronic body **21**, so as to electrically connect the circuit portion **22** and the plurality of first conductors **21***a*. For example, the circuit portion **22** includes at least one passivation layer **220** and conductive traces **221** bonded with the passivation layer **220**, so that the conductive traces **221** are electrically connected to the conductive vias **210** and the plurality of second conductors **22***a*. It should be understood that the structure of the element having the conductive vias **210** has various aspects and the present disclosure is not limited to as such.
[0039] Furthermore, the first conductors **21***a* and the second conductors **22***a* are metal pillars such as copper pillars, and the first protective layer **21***b* is an insulating film or a polyimide (PI) material, so that the first conductors **21***a* are not exposed, and the second protective layer **22***b* is a non-conductive film (NCF) or other materials that are easy to adhere to the insulating layer **24**. For example, the second conductors **22***a* are first formed on the circuit portion **22** of the electronic structure **2***a*, and then the non-conductive film (the second protective layer **22***b*) is adhered. It should be understood that the second protective layer **22***b* may not be formed in the electronic structure **2***a*, and a conventional dispensing (i.e., underfill) process can be used in the subsequent process (in the process shown in FIG. **2**B-**1**), but the second conductors **22***a* adopt the configuration specifications of small pitch, low height and high density, which are not conducive to the capillary flow of the conventional underfill. Therefore, it is preferable to select a non-conductive film as the second protective layer **22***b*.
[0040] As shown in FIG. **2**B-**1**, the electronic structure **2***a* and the second protective layer **22***b* thereon are bonded onto the insulating layer **24**, and at least one reinforcing member **29** is disposed on the insulating layer **24**.
[0041] In an embodiment, the reinforcing member **29** is a metal frame or a non-metal frame, which is fixed on the insulating layer **24** by sticking or other methods. For example, the reinforcing member **29** is annular, as shown in FIG. **2**B-**2**, so as to surround the electronic structure **2***a* and the conductive pillars **23**.
[0042] Furthermore, a width D of the reinforcing member **29** is at least 35 micrometers (μm), and the width D of the reinforcing member **29** is greater than a width R of the conductive pillar **23**. Without affecting the layout of the conductive pillar **23**, the thicker the reinforcing member **29**, the better.
[0043] Please also refer to FIG. **4**A, in other embodiments, a reinforcing member **491** can also be a part of a mask structure **49** and is disposed on a cover body **490**, so that the reinforcing member **491** is erected (e.g., vertically disposed) on the insulating layer **24**, such that the cover body **490**

covers the electronic structure **2***a* and the conductive pillars **23**. For example, the cover body **490** can be in the shape of a rectangular sheet, and the reinforcing member **491** can be in the shape of a wall (as shown in FIG. **4**B), in the shape of a grid (e.g., a reinforcing member **492** as shown in FIG. **4**C), or in the shape of a column (as shown in FIG. **4**D, reinforcing members **493** are provided at the corners of the cover body **490**).

[0044] Therefore, there are various aspects of the reinforcing member **29**, **491**, **492**, **493**, and the present disclosure is not limited to as such.

[0045] As shown in FIG. **2**C-**1**, following the process of FIG. **2**B-**1**, a cladding layer **25** is formed on the insulating layer **24**, so that the cladding layer **25** covers the electronic structure **2***a*, the reinforcing member **29** and the conductive pillars **23**, wherein the cladding layer **25** has a first surface **25***a* and a second surface **25***b* opposing the first surface **25***a*, so that the first protective layer **21***b*, end surfaces of the first conductors **21***a* and end surfaces **23***a* of the conductive pillars **23** are exposed from the first surface **25***a* of the cladding layer **25**, and the cladding layer **25** is bonded onto the insulating layer **24** with the second surface **25***b* thereof.

[0046] In an embodiment, the cladding layer **25** is an insulating material, such as polyimide (PI), dry film, encapsulating colloid such as epoxy resin, or molding compound. For example, the cladding layer **25** can be formed on the insulating layer **24** by liquid compound, injection, lamination, or compression molding.

[0047] Furthermore, the first surface **25***a* of the cladding layer **25** can be flushed with the first protective layer **21***b*, the reinforcing member **29**, the end surfaces **23***a* of the conductive pillars **23** and the end surfaces of the first conductors **21***a* via a leveling process, so that the reinforcing member **29**, the end surfaces **23***a* of the conductive pillars **23** and the end surfaces of the first conductors **21***a* are exposed from the first surface **25***a* of the cladding layer **25**. For example, by a grinding method, the leveling process removes partial materials of the reinforcing member **29**, partial materials of the first protective layer **21***b*, partial materials of the conductive pillar **23**, partial materials of the first conductor **21***a* and partial materials of the cladding layer **25**. It should be understood that, if the process of FIG. **4**A is continued, the cover body **490** can be removed by grinding via the leveling process, so as to present the aspect shown in FIG. **2**C-**1**, so that the reinforcing members **491**, **492**, **493** (like the reinforcing member **29** shown in FIG. **2**C-**1**) are flush with and exposed from the first surface **25***a* of the cladding layer **25**.

[0048] Also, as shown in FIG. **2**C-**2**, the electronic structure **2***a* is defined with a horizontal center line L, so as to divide the cladding layer **25** into a first area A**1** and a second area A**2** opposite to each other, so that when the number of the conductive pillars **23** in the first area A**1** is greater than the number of the conductive pillars **23** in the second area A**2**, a width D**2** of the reinforcing member **29** corresponding to the second area A**2** is greater than a width D**1** of the reinforcing member **29** corresponding to the first area A**1**, so as to balance volume of copper in the first area A**1** and the second area A**2**.

[0049] In addition, the reinforcing member **29** may have at least one guide hole **290** for the cladding layer **25** to flow into the reinforcing member **29** through the guide hole **290** to cover the electronic structure **2***a* and the conductive pillars **23**. For example, the reinforcing member **29** is configured with a plurality of guide holes **290** to form a discontinuous ring, so as to facilitate the flow of the adhesive material of the cladding layer **25**. Further, the position of the guide hole **290** is deviated from the horizontal center line L of the electronic structure **2***a*, so as to avoid the problem that the adhesive material of the cladding layer **25** directly collides with the electronic structure **2***a* and causes the electronic structure **2***a* to be broken.

[0050] As shown in FIG. **2**D, a circuit structure **20** is formed on the first surface **25***a* of the cladding layer **25**, so that the circuit structure **20** is electrically connected to the plurality of conductive pillars **23** and the plurality of first conductors **21***a*, and the circuit structure **20** is free from being electrically connected to the reinforcing member **29**.

[0051] In an embodiment, the circuit structure **20** includes at least one insulating layer **200** and at least one redistribution layer (RDL) **201** disposed on the insulating layer **200**, wherein the outermost insulating layer **200** can be used as a solder mask layer, and the outermost redistribution layer **201** is exposed from the solder mask layer to serve as electrical contact pads **202**, such as micro pads (commonly known as u-pads).

[0052] Furthermore, the material for forming the redistribution layer **201** is copper, and the material for forming the insulating layer **200** is a dielectric material such as polybenzoxazole (PBO), polyimide (PI), prepreg (PP) and the like, or a solder-proof material such as solder mask, ink (e.g., graphite) and the like.

[0053] As shown in FIG. **2**E, a plurality of electronic elements **26** are disposed on the circuit structure **20**, and then a package layer **28** is used to cover the electronic elements **26**.

[0054] In an embodiment, the electronic element **26** is an active element, a passive element, or a combination of the active element and the passive element, where the active element is such as a semiconductor chip, and the passive element is such as a resistor, a capacitor, or an inductor. In one embodiment, the electronic element **26** is, for example, a semiconductor chip such as a graphics processing unit (GPU), a high bandwidth memory (HBM), etc., and the present disclosure is not limited to as such. Further, the electronic structure **2***a* is used as a bridge element (e.g., a bridge die), which is electrically connected to the circuit structure **20** via the first conductors **21***a* to electrically bridge at least two electronic elements **26**.

[0055] Furthermore, an active surface **26***a* of the electronic element **26** has a plurality of electrode pads **260** to electrically connect the electrical contact pads **202** via a plurality of conductive bumps **261** containing solder material, and the package layer **28** can cover the electronic elements **26** and the conductive bumps **261** at the same time. In an embodiment, an under bump metallurgy (UBM) layer (not shown) may be formed on the electrical contact pads **202** to facilitate bonding with the conductive bumps **261**.

[0056] In addition, the package layer **28** is an insulating material, such as polyimide (PI), dry film, encapsulating colloid such as epoxy resin, or molding compound, which can be formed on the circuit structure **20** by lamination or molding. It should be understood that the material for forming the package layer **28** can be the same as or different from the material for forming the cladding layer **25**.

[0057] In addition, an underfill **262** can also be first formed between the electronic element **26** and the circuit structure **20** to cover the conductive bumps **261**, and then the package layer **28** can be formed to cover the underfill **262** and the electronic element **26**.

[0058] As shown in FIG. **2**F, the carrier **9** and the release layer **90** thereon are removed, and then the metal layer **91** is removed to expose the insulating layer **24** and the other end surfaces **23***b* of the conductive pillars **23**.

[0059] In an embodiment, when the release layer **90** is peeled off, the metal layer **91** is used as a barrier to avoid damage to the insulating layer **24**, and after removing the carrier **9** and the release layer **90** thereon, the metal layer **91** is removed by etching. At this time, the conductive pillars **23** are exposed from the insulating layer **24**, and the electronic structure **2***a* and the reinforcing member **29** are free from being exposed from the insulating layer **24**.

[0060] As shown in FIG. **2**G, a plurality of conductive elements **27** are formed on the insulating layer **24**, and the conductive elements **27** are electrically connected to the plurality of conductive pillars **23** and the plurality of second conductors **22***a*.

[0061] In an embodiment, an opening process is performed on the insulating layer **24**, so that the second conductors **22***a* are exposed from the insulating layer **24** for bonding the plurality of conductive elements **27**. For example, a plurality of openings are formed on the insulating layer **24** by a laser method, so that the second conductors **22***a* are exposed from the openings, such that the plurality of conductive elements **27** are formed in the plurality of openings to electrically connect the second conductors **22***a*, wherein each of the conductive elements **27** includes a metal body (such as UBM or circuit) **270** and a copper pillar **271** bonded to the metal body **270** to form a

solder material **27***a* such as a solder bump or a solder ball on the end surface of the copper pillar **271**.
[0062] It should be understood that, when the number of contacts (e.g., input/output or I/O) is insufficient (e.g., if the number of the conductive elements **27** can no longer meet the product requirements), a build-up operation can still be performed on the insulating layer **24** by the RDL process.
[0063] As shown in FIG. **2**H-**1**, a singulation process is performed along a cutting path S shown in FIG. **2**G to obtain a plurality of electronic packages **2**, wherein the reinforcing member **29** is located on the cutting path S. Therefore, after the singulation process, the cladding layer **25** will be formed with a plurality of side surfaces **25***c* adjacent to the first surface **25***a* and the second surface **25***b*, and the reinforcing member **29** is disposed on the side surface **25***c* of the cladding layer **25** and is exposed from the cladding layer **25**.
[0064] In an embodiment, a leveling process, such as grinding, can be used to remove partial materials of the package layer **28**, so that the upper surface of the package layer **28** is flush with the upper surface of the electronic element **26**, such that the electronic element **26** is exposed from the package layer **28**.
[0065] Furthermore, the reinforcing member **29** can be arranged at any position on the insulating layer **24** according to requirements. For example, as shown in FIG. **2**H-**2**, at least one reinforcing member **39** is disposed between the conductive pillar **23** and the electronic structure **2***a* and embedded in the cladding layer **25**.
[0066] Also, in the subsequent process, the electronic package **2** can be disposed on a package substrate **3** via the conductive elements **27**, as shown in FIG. **3**, and a ball-placement process is performed on the underside of the package substrate **3** to form a plurality of solder balls **30** for the package substrate **3** to be disposed on a circuit board (not shown) with the plurality of solder balls **30**.
[0067] In addition, the package substrate **3** can be provided with a functional member **31** (such as a metal frame) according to requirements, so as to eliminate the problem of stress concentration and prevent the package substrate **3** from warping.
[0068] Therefore, in the manufacturing method of the present disclosure, the reinforcing members **29**, **39** are disposed on the insulating layer **24** before the cladding layer **25** is formed, so as to enhance the structural strength of the electronic package **2** during the manufacturing process. Therefore, compared with the prior art, the present disclosure can effectively avoid the problem of warpage of the electronic package **2** before and after the manufacturing process.
[0069] Furthermore, the electronic structure **2***a* is used as a bridge element (e.g., a bridge die) to directly electrically conduct at least two active chips (electronic elements **26**) above, so as to shorten the electrical path, and the pitch between the contacts (I/O) or the electrical contact pads **202** can be effectively reduced according to requirements, and the number of layers of the redistribution layer **201** for electrical connection between the upper and lower layers of the circuit structure **20** can also be reduced to increase the process yield.
[0070] In addition, the electronic structure **2***a* has conductive vias **210**, so that part of the electrical path (such as power) can be directly transmitted up and down through the electronic structure **2***a* to the required place (such as the package substrate **3** or the electronic element **26**), so as to shorten the electrical path and improve the electrical performance.
[0071] The present disclosure also provides an electronic package **2**, which comprises: a cladding layer **25**, an electronic structure **2***a*, a plurality of conductive pillars **23**, at least one reinforcing member **29**, **39**, a circuit structure **20** and a plurality of electronic elements **26**.
[0072] The cladding layer **25** has a first surface **25***a* and a second surface **25***b* opposing the first surface **25***a*.
[0073] The electronic structure **2***a* is embedded in the cladding layer **25**.
[0074] The conductive pillars **23** are embedded in the cladding layer **25**.

[0075] The reinforcing members **29**, **39** are bonded to the cladding layer **25**.
[0076] The circuit structure **20** is formed on the first surface **25***a* of the cladding layer **25** and is electrically connected to the plurality of conductive pillars **23** and the electronic structure **2***a*, and the circuit structure **20** is free from being electrically connected to the reinforcing members **29**, **39**.
[0077] The electronic elements **26** are disposed on the circuit structure **20** and are electrically connected to the circuit structure **20**, so that the electronic structure **2***a* is used as a bridge die, such that the bridge die electrically bridges at least two of the plurality of electronic elements **26** via the circuit structure **20**.
[0078] In one embodiment, the reinforcing members **29**, **39** are metal frames or non-metal frames.
[0079] In one embodiment, widths D, D**1**, D**2** of the reinforcing members **29**, **39** are at least **35** micrometers.
[0080] In one embodiment, the widths D, D**1**, D**2** of the reinforcing members **29**, **39** are greater than or equal to a width R of each of the conductive pillars **23**.
[0081] In one embodiment, the reinforcing members **29**, **39** are annular.
[0082] In one embodiment, the reinforcing members **29**, **39** surround the electronic structure **2***a*.
[0083] In one embodiment, the reinforcing members **29**, **39** have at least one guide hole **290**.
[0084] In one embodiment, the electronic structure **2***a* is defined with a horizontal center line L, so that the cladding layer **25** is divided into a first area A**1** and a second area A**2** opposite to each other, so that the number of the conductive pillars **23** in the first area A**1** defined by the horizontal center line L is greater than the number of the conductive pillars **23** in the second area A**2** defined by the horizontal center line L, such that the width D**2** of the reinforcing member **29** corresponding to the second area A**2** divided by the horizontal center line L is greater than the width D**1** of the reinforcing member **29** corresponding to the first area A**1** divided by the horizontal center line L.
[0085] In one embodiment, the electronic package **2** further includes a plurality of conductive elements **27** disposed on the second surface **25***b* of the cladding layer **25**, where the plurality of conductive elements **27** are electrically connected to the plurality of conductive pillars **23** and are free from being electrically connected to the reinforcing members **29**, **39**.
[0086] To sum up, the electronic package of the present disclosure and the manufacturing method thereof are designed to enhance the structural strength of the electronic package by means of the design of the reinforcing member. Therefore, the present disclosure can effectively avoid the warping problem of the electronic package.
[0087] The foregoing embodiments are provided for the purpose of illustrating the principles and effects of the present disclosure, rather than limiting the present disclosure. Anyone skilled in the art can modify and alter the above embodiments without departing from the spirit and scope of the present disclosure. Therefore, the scope of protection with regard to the present disclosure should be as defined in the accompanying claims listed below.

Claims

1. An electronic package, comprising: a cladding layer having a first surface and a second surface opposing the first surface; an electronic structure embedded in the cladding layer; at least one reinforcing member bonded to the cladding layer; and a plurality of electronic elements, wherein the electronic structure acts as a bridge element to electrically bridge at least two of the plurality of electronic elements.
2. The electronic package of claim 1, wherein the reinforcing member is a metal frame.
3. The electronic package of claim 1, wherein the reinforcing member is a frame made of non-metallic material.
4. The electronic package of claim 1, wherein the reinforcing member has a width of at least 35 micrometers.
5. The electronic package of claim 1, further comprising a plurality of conductive pillars embedded

in the cladding layer.

6. The electronic package of claim 5, further comprising a circuit structure formed on the first surface of the cladding layer and electrically connected to the plurality of conductive pillars and the electronic structure, wherein the circuit structure is free from being electrically connected to the reinforcing member.

7. The electronic package of claim 6, wherein the plurality of electronic elements are disposed on and electrically connected to the circuit structure, and wherein the electronic structure electrically bridges at least two of the plurality of electronic elements via the circuit structure.

8. The electronic package of claim 5, wherein the reinforcing member has a width greater than or equal to a width of each of the conductive pillars.

9. The electronic package of claim 1, wherein the reinforcing member is annular.

10. The electronic package of claim 1, wherein the reinforcing member surrounds the electronic structure.

11. The electronic package of claim 1, wherein the reinforcing member has at least one guide hole.

12. The electronic package of claim 5, wherein the electronic structure is defined with a horizontal center line to divide the cladding layer into a first area and a second area opposing the first area, such that a number of the conductive pillars in the first area is greater than a number of the conductive pillars in the second area, and a width of the reinforcing member corresponding to the second area is greater than a width of the reinforcing member corresponding to the first area.

13. The electronic package of claim 5, further comprising a plurality of conductive elements disposed on the second surface of the cladding layer, wherein the plurality of conductive elements are electrically connected to the plurality of conductive pillars and are free from being electrically connected to the reinforcing member.